

[Continued from §34 of Paper 141. — The present file replaces the corresponding section of the 251–300 chunk with a higher-fidelity rendering of the covariant tables on pages 260–272 and of the numerical ad-, ae-, ..., ai-tables of Paper 142, pp. 276–281.]

I have worked out the example in detail as a specimen of the most convenient method for the actual calculation of more complicated covariants¹.

35. The number of terms of the degree θ and of the weight q is obviously equal to the number of ways in which q can be made up as a sum of θ terms with the elements $(0, 1, 2, \dots, m)$, a number which is equal to the coefficient of $x^q z^\theta$ in the development of

$$\frac{1}{(1-z)(1-xz)(1-x^2z)\cdots(1-x^mz)},$$

and the number of the aszygetic covariants of any particular degree for the quantic $(* \mid x, y)^m$ can therefore be determined by means of this development. In the case of a cubic, for example, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)},$$

which is equal to

$$1 + (1 + x + x^2 + x^3)z + (1 + x + 2x^2 + 2x^3 + 2x^4 + x^5 + x^6)z^2 + \&c.,$$

where the coefficients are given by the following table; on account of the symmetry, the series of coefficients for each power of z is continued only to the middle term or middle of the series.

[Figure: Triangular table of partition counts for the cubic, rows (0)–(6), entries continued only to the middle: row (0) is 1; row (1) is 1, 1; row (2) is 1, 1, 2; row (3) is 1, 1, 2, 3; row (4) is 1, 1, 2, 3, 4, 5; row (5) is 1, 1, 2, 3, 4, 5, 6; row (6) is 1, 1, 2, 3, 4, 5, 7, 8 (mirrored about the middle).]

36. and from this, by subtracting from each coefficient the coefficient which immediately precedes it, we form the table

[Figure: Triangular difference table for the cubic, rows (0)–(6); entries are the successive differences. Row (6) reads 1, 0, 1, 1, 1, 1, 2, 1, 0, mirrored.]

The successive lines fix the number and character of the covariants of the degrees 0, 1, 2, 3, &c. The line (0), if this were to be interpreted, would show that there is a single covariant of the degree 0; this covariant is of course merely the absolute constant unity, and may be excluded. The line (1) shows that there is a single covariant of the degree 1, viz. a covariant of the order 3; this is the cubic itself, which I represent by U . The line (2) shows that there are two aszygetic covariants of the degree 2, viz. one of the order 6, this is merely U^2 , and one of the order 2, this I represent by H . The line (3) shows that there are three aszygetic covariants of the degree 3, viz. one of the order 9, this is U^3 ; one of the order 5, this is UH ;

¹Note added Feb. 7, 1856.—The following method for the calculation of an invariant or of the leading coefficient of a covariant, is easily seen to be identical in principle with that given in the text. Write down all the terms of the weight next inferior to that of the invariant or leading coefficient, and operate on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{\partial}{\partial a} + 2 \text{ ind. } c \cdot \frac{\partial}{\partial b} + \cdots + (m-1) \text{ ind. } k \cdot \frac{\partial}{\partial j},$$

where we are first to multiply by the fraction, rejecting negative powers, and then by the index of the proper letter in the term so obtained. Equating the results to zero, we obtain equations between the terms of the invariant or leading coefficient, and replacing in these equations each term by its numerical coefficient in the invariant or leading coefficient, we have the equations of connexion of these numerical coefficients. Thus, for the discriminant of a cubic, the terms of the next inferior weight are a^2cd , abc^2 , ab^2d , b^3c , and operating on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{\partial}{\partial a} + 2 \text{ ind. } c \cdot \frac{\partial}{\partial b} + 3 \text{ ind. } d \cdot \frac{\partial}{\partial c},$$

we find

$$\begin{aligned} a^2cd &: abcd + 4abcd + 6ac^3 \\ abc^2 &: 4b^3c + 4abcd + 6ac^3 \\ ab^2d &: 2b^3c + 4b^3d \\ b^3c &: 6b^2c^2 \end{aligned}$$

and equating the horizontal lines to zero, and assuming $a^2d^2 = 1$, we have $a^2d^2 = 1$, $abcd = -6$, $ac^3 = 4$, $b^3d = 4$, $b^2c^2 = -3$, or the value of the discriminant is that given in the text.

and one of the order 3, this I represent by Φ . The line (4) shows that there are five asyzygetic covariants of the degree 4, viz. one of the order 12, this is U^4 ; one of the order 8, this is U^2H ; one of the order 6, this is H^2 ; one of the order 4, this is $U\Phi$; and one of the order 0, i.e. an invariant, this I represent by ∇ . The line (5) shows that there are six asyzygetic covariants of the degree 5, viz. one of the order 15, this is U^5 ; one of the order 11, this is U^3H ; one of the order 9, this is $U^2\Phi$; one of the order 7, this is UH^2 ; one of the order 5, this is $H\Phi$; and one of the order 3, this is $U\nabla$. The line (6) shows that there are 8 asyzygetic covariants of the degree 6, viz. one of the order 18, this is U^6 ; one of the order 14, this is U^4H ; one of the order 12, this is $U^3\Phi$; one of the order 10, this is U^2H^2 ; one of the order 8, this is $UH\Phi$; two of the order 6 (i.e. the three covariants H^3 , Φ^2 and ∇U^2 are not asyzygetic, but are connected by a single linear equation or syzygy); and one of the order 2, this is ∇H . We are thus led to the irreducible covariants U, H, Φ, ∇ connected by a linear equation or syzygy between H^3 , Φ^2 and ∇U^2 , and this is in fact the complete system of irreducible covariants; ∇ is therefore the only invariant.

36.² The asyzygetic covariants are of the form $U^pH^q\nabla^r$, or else of the form $\Phi U^pH^q\nabla^r$; and since U, H, ∇ are of the degrees 1, 2, 4 respectively, and Φ is of the degree 3, the number of asyzygetic covariants of the degree m of the first form is equal to the coefficient of x^m in $1 \div (1-x)(1-x^2)(1-x^4)$, and the number of the asyzygetic covariants of the degree m of the second form is equal to the coefficient of x^m in $x^3 \div (1-x)(1-x^2)(1-x^4)$. Hence the total number of asyzygetic covariants is equal to the coefficient of x^m in $(1+x^3) \div (1-x)(1-x^2)(1-x^4)$, or what is the same thing, in

$$\frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)},$$

and conversely, if this expression for the number of the asyzygetic covariants of the degree m were established independently, it would follow that the irreducible invariants were four in number, and of the degrees 1, 2, 3, 4 respectively, but connected by an equation of the degree 6. As regards the invariants, every invariant is of the form ∇^p , i.e. the number of asyzygetic invariants of the degree m is equal to the coefficient of x^m in $\frac{1}{1-x^4}$, and conversely, from this expression it would follow that there was a single irreducible invariant of the degree 4.

37. In the case of a quartic, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)(1-x^4z)},$$

and the coefficients are given by the table.

[Figure: Triangular coefficient table for the quartic, rows (0)–(6). Last row (6) reads 1, 1, 2, 3, 5, 6, 9, 10, 13, 14, 16, 16, 18, mirrored.]

and subtracting from each coefficient the coefficient immediately preceding it, we have the table

[Figure: Triangular difference table for the quartic, rows (0)–(6).]

the examination of which will show that we have for the quartic the following irreducible covariants, viz. the quartic itself U ; an invariant of the degree 2, which I represent by I ; a covariant of the order 4 and of the degree 2, which I represent by H ; an invariant of the degree 3, which I represent by J ; and a covariant of the order 6 and the degree 3, which I represent by Φ ; but that the irreducible covariants are connected by an equation of the degree 6, viz. there is a linear equation or syzygy between Φ^2 , I^2H^2 , IJH^2U , JH^2U^2 and J^2U^2 ; this is in fact the complete system of the irreducible covariants of the quartic: the only irreducible invariants are the invariants I, J .

38. The asyzygetic covariants are of the form $U^pI^qH^rJ^s$, or else of the form $\Phi U^pI^qH^rJ^s$, and the number of the asyzygetic covariants of the degree m is equal to the coefficient of x^m in $(1+x^3) \div (1-x)(1-x^2)^2(1-x^3)$, or what is the same thing, in

$$\frac{1-x^6}{(1-x)(1-x^2)^2(1-x^3)^2},$$

and the asyzygetic invariants are of the form I^pJ^q , and the number of the asyzygetic invariants of the degree m is equal to the coefficient of x^m in $1 \div (1-x^2)(1-x^3)$. Conversely, if these formulæ were

²Two consecutive paragraphs are numbered **36** in the original.

established, the preceding results as to the form of the system of the irreducible covariants or of the irreducible invariants, would follow.

39. In the case of a quintic, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)(1-x^4z)(1-x^5z)},$$

and the coefficients are given by the table

[Figure: Triangular coefficient table for the quintic, rows (0)–(5).]

and subtracting from each coefficient the one which immediately precedes it, we have the table

[Figure: Triangular difference table for the quintic, rows (0)–(5).]

We thus obtain the following irreducible covariants, viz.:

Of the degree 1; a single covariant of the order 5, this is the quintic itself.

Of the degree 2; two covariants, viz. one of the order 6, and one of the order 2.

Of the degree 3; three covariants, viz. one of the order 9, one of the order 5, and one of the order 3.

Of the degree 4; three covariants, viz. one of the order 6, one of the order 4, and one of the order 0 (an invariant).

Of the degree 5; three covariants, viz. one of the order 7, one of the order 3, and one of the order 1 (a linear covariant).

These covariants are connected by a single syzygy of the degree 5 and of the order 11; in fact, the table shows that there are only two asyzygetic covariants of this degree and order; but we may, with the above-mentioned irreducible covariants of the degrees, 1, 2, 3 and 4, form three covariants of the degree 5 and the order 11; there is therefore a syzygy of this degree and order.

40. I represent the number of ways in which q can be made up as a sum of m terms with the elements $0, 1, 2, \dots, m$, each element being repeatable an indefinite number of times by the notation

$$P(0, 1, 2, \dots, m)^\theta q,$$

and I write for shortness

$$P'(0, 1, 2, \dots, m)^\theta q = P(0, 1, 2, \dots, m)^\theta q - P(0, 1, 2, \dots, m)^\theta (q-1).$$

Then for a quantic of the order m , the number of asyzygetic covariants of the degree θ and of the order μ is

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - \mu).$$

In particular, the number of asyzygetic invariants of the degree θ is

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta.$$

To find the total number of the asyzygetic covariants of the degree θ , suppose first that $m\theta$ is even; then, giving to μ the successive values $0, 2, 4, \dots, m\theta$, the required number is

$$\begin{aligned} & P(\tfrac{1}{2}m\theta) - P(\tfrac{1}{2}m\theta - 1) \\ & + P(\tfrac{1}{2}m\theta - 1) - P(\tfrac{1}{2}m\theta - 2) \\ & \vdots \\ & + P(2) - P(1) \\ & + P(1) \\ & = P(\tfrac{1}{2}m\theta), \end{aligned}$$

i.e. when $m\theta$ is even, the number of the asyzygetic covariants of the degree θ is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta;$$

and similarly, when $m\theta$ is odd, the number of the asyzygetic covariants of the degree θ is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1).$$

But the two formulae may be united into a single formula; for when $m\theta$ is odd $\frac{1}{2}m\theta$ is a fraction, and therefore $P(\frac{1}{2}m\theta)$ vanishes, and so when $m\theta$ is even $\frac{1}{2}(m\theta - 1)$ is a fraction, and $P\frac{1}{2}(m\theta - 1)$ vanishes; we have thus the theorem, that for a quantic of the order m :

The number of the asyzygetic covariants of the degree θ is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta + P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1).$$

41. The functions $P(\frac{1}{2}m\theta)$, &c. may, by the method explained in my “Researches on the Partition of Numbers,” [140], be determined as the coefficients of x^θ in certain functions of x ; I have calculated the following particular cases:

Putting, for shortness,

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta = \text{coefficient of } x^\theta \text{ in } \phi m,$$

then

$$\begin{aligned}\phi 2 &= \frac{1}{1 - x^2}, \\ \phi 3 &= \frac{1}{1 - x^4}, \\ \phi 4 &= \frac{1}{(1 - x^2)(1 - x^3)}, \\ \phi 5 &= \frac{1 - x^8 + x^{18}}{(1 - x^4)(1 - x^8)(1 - x^{12})}, \\ \phi 6 &= \frac{(1 - x)(1 + x - x^2 - x^4 - x^5 + x^7 + x^8)}{(1 - x^2)^2(1 - x^3)(1 - x^4)(1 - x^5)}, \\ \phi 7 &= \frac{1 - x^4 + 2x^8 - x^{10} + 5x^{12} + 2x^{14} + 6x^{16} + 2x^{18} + 5x^{20} - x^{22} + 2x^{24} - x^{28} + x^{32}}{(1 - x^4)(1 - x^8)(1 - x^{12})(1 - x^{14})(1 - x^{18})}, \\ \phi 8 &= \frac{(1 - x)(1 + x - x^2 - x^4 + x^7 + x^8 + x^9 + x^{10} - x^{11} + x^{13} + x^{14})}{(1 - x^2)^2(1 - x^3)^2(1 - x^4)(1 - x^5)(1 - x^6)},\end{aligned}$$

and putting

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta = \text{coefficient of } x^\theta \text{ in } \psi m,$$

then

$$\begin{aligned}\psi 2 &= \frac{1}{(1 - x)(1 - x^2)}, \\ \psi 3 &= \frac{1 + x^3}{(1 - x^2)(1 - x^4)}, \\ \psi 4 &= \frac{1 - x + x^2}{(1 - x)^2(1 - x^2)(1 - x^3)}, \\ \psi 5 &= \frac{1 + x^4 + 6x^6 + 9x^7 + 12x^8 + 9x^9 + 6x^{10} + x^{12} + x^{16}}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)},\end{aligned}$$

and putting

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1) = \text{coefficient of } x^\theta \text{ in } \chi m,$$

then

$$\begin{aligned}\chi 3 &= \frac{x + x^3}{(1 - x^2)^2(1 - x^3)}, \\ \chi 5 &= \frac{x + 4x^2 + 8x^3 + 10x^4 + 10x^5 + 8x^6 + 4x^7 + x^8}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)}.\end{aligned}$$

And from what has preceded, it appears that for a quantic of the order m , the number of asyzygetic covariants of the degree θ is for m even, coefficient x^θ in ψm , and for m odd, coefficient x^θ in $(\psi m + \chi m)$;

and that the number of asyzygetic invariants of the degree θ is coefficient x^θ in ϕm . Attending first to the invariants:

42. For a quadric, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{1-x^2},$$

which leads to the conclusion that there is a single irreducible invariant of the degree 2.

43. For a cubic, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{1-x^4},$$

i.e. there is a single irreducible invariant of the degree 4.

44. For a quartic, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{(1-x^2)(1-x^3)},$$

i.e. there are two irreducible invariants of the degrees 2 and 3 respectively.

45. For a quintic, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^8+x^{18}}{(1-x^4)(1-x^8)(1-x^{12})}.$$

The numerator is the irreducible factor of $1-x^{36}$, i.e. it is equal to $(1-x^{36})(1-x^4) \div (1-x^{12})(1-x^9)$; and substituting this value, the number becomes

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^{36}}{(1-x^4)(1-x^8)(1-x^{12})(1-x^{18})},$$

i.e. there are in all four irreducible invariants, which are of the degrees 4, 8, 12 and 18 respectively; but these are connected by an equation of the degree 36, i.e. the square of the invariant of the degree 18 is a rational and integral function of the other three invariants; a result, the discovery of which is due to M. Hermite.

46. For a sextic, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{(1-x)(1+x-x^2-x^4-x^5+x^7+x^8)}{(1-x^2)^2(1-x^3)(1-x^4)(1-x^5)};$$

the second factor of the numerator is the irreducible factor of $1-x^{30}$, i.e. it is equal to $(1-x^{30})(1-x^5)(1-x^3)(1-x^2) \div (1-x^{15})(1-x^{10})(1-x^6)(1-x)$; and substituting this value, the number becomes

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^{30}}{(1-x^2)(1-x^4)(1-x^6)(1-x^{10})(1-x^{15})},$$

i.e. there are in all five irreducible invariants, which are of the degrees 2, 4, 6, 10 and 15 respectively; but these are connected by an equation of the degree 30, i.e. the square of the invariant of the degree 15 is a rational and integral function of the other four invariants.

47. For a septic, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^4+2x^8-x^{10}+5x^{12}+2x^{14}+6x^{16}+2x^{18}+5x^{20}-x^{22}+2x^{24}-x^{28}+x^{32}}{(1-x^4)(1-x^8)(1-x^{12})(1-x^{14})(1-x^{18})};$$

the numerator is equal to

$$(1-x^4)(1-x^8)^{-1}(1-x^{10})(1-x^{12})^{-1}(1-x^{14})^{-1}\dots,$$

where the series of factors does not terminate; hence [incorrect, see p. 253] the number of irreducible invariants is infinite; substituting the preceding value, the number of asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } (1-x^4)^{-1}(1-x^8)^{-3}(1-x^{12})^{-6}(1-x^{14})^{-4}\dots.$$

The first four indices give the number of irreducible invariants of the corresponding degrees, i.e. there are 1, 3, 6 and 4 irreducible invariants of the degrees 4, 8, 12 and 14 respectively, but there is no reason to believe that the same thing holds with respect to the indices of the subsequent terms. To verify this it is to be remarked, that there are 1, 4, 10 and 4 asyzygetic invariants of the degrees in question respectively; there is therefore one irreducible invariant of the degree 4; calling this X_4 , there is only one composite invariant of the degree 8, viz. X_4^2 ; there are therefore three irreducible invariants of this degree, say X_8 , X'_8 , X''_8 . The composite invariants of the degree 12 are four in number, viz. X_4^3 , X_4X_8 , $X_4X'_8$, $X_4X''_8$, and these cannot be connected by any syzygy, for if they were so, X_4^2 , X_8 , X'_8 , X''_8 would be connected by a syzygy, or there would be less than 3 irreducible invariants of the degree 8. Hence there are precisely 6 irreducible invariants of the degree 12. And since the irreducible invariants of the degrees 4, 8 and 12 do not give rise to any composite invariant of the degree 14, there are precisely 4 irreducible invariants of the degree 14.

48. For an octavic, the number of the asyzygetic invariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{(1-x)(1+x-x^2-x^4+x^7+x^8+x^9+x^{10}-x^{11}+x^{13}+x^{14})}{(1-x^2)^2(1-x^3)^2(1-x^4)(1-x^5)(1-x^6)},$$

and the second factor of the numerator is

$$(1-x)^{-1}(1-x^2)^{-1}(1-x^3)^{-1}(1-x^4)^{-1}(1-x^5)^{-1}(1-x^6)^{-1}(1-x^{10})^{-1}(1-x^{12})(1-x^{14})(1-x^{15})\dots,$$

where the series of factors does not terminate; hence [incorrect] the number of irreducible invariants is infinite. Substituting the preceding value, the number of the asyzygetic invariants of the degree θ is

$$\text{coeff. of } x^\theta \text{ in } (1-x^2)^{-1}(1-x^3)^{-1}(1-x^4)^{-1}(1-x^5)^{-1}(1-x^6)^{-1}(1-x^7)^{-2}(1-x^8)^{-3}(1-x^9)^{-5}(1-x^{10})^{-7}\dots$$

There is certainly one, and only one irreducible invariant for each of the degrees 2, 3, 4, 5 and 6 respectively; but the formula does not show the number of the irreducible invariants of the degrees 7, &c.; in fact, representing the irreducible invariants of the degrees 2, 3, 4, 5 and 6 by X_2 , X_3 , X_4 , X_5 , X_6 , these give rise to 3 composite invariants of the degree 7, viz. $X_2X_2X_3$, X_3X_4 , X_2X_5 , which may or may not be connected by a syzygy; if they are not connected by a syzygy, there will be a single irreducible invariant of the degree 7; but if they are connected by a syzygy, there will be two irreducible invariants of the degree 7; it is useless at present to pursue the discussion further.

Considering next the covariants,

49. For a quadric, the number of asyzygetic covariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{(1-x)(1-x^2)},$$

i.e. there are two irreducible covariants of the degrees 1 and 2 respectively; these are of course the quadric itself and the invariant.

50. For a cubic, the number of the asyzygetic covariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{(1+x)(1+x^3)}{(1-x^2)^2(1-x^4)}.$$

The first factor of the numerator is the irreducible factor of $1-x^2 = (1-x^2) \div (1-x)$, and the second factor of the numerator is the irreducible factor of $1-x^6 = (1-x^6) \div (1-x^2)$; substituting these values, the number is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^6}{(1-x)(1-x^2)(1-x^3)(1-x^4)},$$

i.e. there are 4 irreducible covariants of the degrees 1, 2, 3, 4 respectively; but these are connected by an equation of the degree 6; the covariant of the degree 1 is the cubic itself U , the other covariants are the covariants already spoken of and represented by the letters H , Φ and ∇ respectively (H is of the degree 2 and the order 3, Φ of the degree 3 and the order 3, and ∇ is of the degree 4 and the order 0, i.e. it is an invariant).

51. For a quartic, the number of the asyzygetic covariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x+x^2}{(1-x)^2(1-x^2)(1-x^3)};$$

the numerator of which is the irreducible factor of $1 - x^6$, i.e. it is equal to $(1 - x^6)(1 - x) \div (1 - x^2)(1 - x^3)$. Making this substitution, the number is

$$\text{coefficient of } x^\theta \text{ in } \frac{1 - x^6}{(1 - x)(1 - x^2)^2(1 - x^3)^2},$$

i.e. there are five irreducible covariants, one of the degree 1, two of the degree 2, and two of the degree 3, but these are connected by an equation of the degree 6. The irreducible covariant of the degree 1 is of course the quartic itself U , the other irreducible covariants are those already spoken of and represented by I, H, J, Φ respectively (I is of the degree 2 and the order 0, and J is of the degree 3 and the order 0, i.e. I and J are invariants, H is of the degree 2 and the order 4, Φ is of the degree 3 and the order 6).

52. For a quintic, the number of irreducible covariants of the degree θ is

$$\text{coefficient of } x^\theta \text{ in } \frac{1 + x + x^2 + 4x^3 + 6x^4 + 8x^5 + 9x^6 + 10x^7 + 12x^8 + 10x^9 + 9x^{10} + 8x^{11} + 6x^{12} + 4x^{13} + x^{14} + x^{15} + x^{16}}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)};$$

the numerator of which is

$$(1 + x)^2(1 + x - x^2 - x^4 + 2x^5 + 3x^6 + 5x^7 + 4x^8 + 3x^9 + 2x^{10} - x^{12} - x^{13} + x^{14}).$$

The first factor is $(1 - x^2)^2 \div (1 - x)^2$, the second factor is

$$(1 - x^2)^3(1 - x^3)^2(1 - x^4)^3(1 - x^5)^4(1 - x^7)^{-5}(1 - x^8)^{-7} \dots,$$

which does not terminate; hence [incorrect] the number of irreducible covariants is infinite. Substituting the preceding values, the expression for the number of the asyzygetic covariants of the degree θ is

$$\text{coeff. of } x^\theta \text{ in } (1 - x)^{-1}(1 - x^2)^{-2}(1 - x^3)^{-3}(1 - x^4)^{-3}(1 - x^5)^{-3}(1 - x^6)^{-4}(1 - x^7)^{-5}(1 - x^8)^{-7} \dots,$$

which agrees with a previous result: the numbers of irreducible covariants for the degrees 1, 2, 3, 4 are 1, 2, 3 and 3 respectively, and for the degree 5, the number of irreducible covariants is three, but there is one syzygy between the composite covariants of the degree in question; the difference $3 - 1 = 2$ is the index taken with its sign reversed of the factor $(1 - x^5)^{-3}$.

53. I consider a system of the asyzygetic covariants of any particular degree and order of a given quantic; the system may of course be replaced by a system the terms of which are any linear functions of those of the original system, and it is necessary to inquire what covariants ought to be selected as most proper to represent the system of asyzygetic covariants; the following considerations seem to me to furnish a convenient rule of selection. Let the literal parts of the terms which enter into the coefficients of the highest power of x or leading coefficients be represented by $M_\alpha, M_\beta, M_\gamma, \dots$ these quantities being arranged in the natural or alphabetical order; the first in order of these quantities M , which enters into the leading coefficient of a particular covariant, may for shortness be called the leading term of such covariant, and a covariant the leading term of which is posterior in order to the leading term of another covariant, may be said to have a lower leading term.

It is clear, that by properly determining the multipliers of the linear functions we may form a covariant the leading term of which is lower than the leading term of any other covariant (the definition implies that there is but one such covariant); call this Θ_1 . We may in like manner form a covariant such that its leading term is lower than the leading term of every other covariant except Θ_1 ; or rather we may form a system of such covariants, since if Θ_2 be a covariant having the property in question, $\Theta_2 + k\Theta_1$ will have the same property, but k may be determined so that the covariant shall not contain the leading term of Θ_1 , i.e. we may form a covariant Θ_2 such that its leading term is lower than the leading term of every other covariant excepting Θ_1 , and that the leading term of Θ_1 does not enter into Θ_2 ; and there is but one such covariant, Θ_2 . Again, we may form a covariant Θ_3 such that its leading term is lower than the leading term of every other covariant excepting Θ_1 and Θ_2 , and that the leading terms of Θ_1 and Θ_2 do not either of them enter into Θ_3 ; and there is but one such covariant, Θ_3 . And so on, until we arrive at a covariant the leading term of which is higher than the leading terms of the other covariants, and which does not contain the leading terms of the other covariants. We have thus a series of covariants $\Theta_1, \Theta_2, \Theta_3, \&c.$ containing the proper number of terms, and which covariants may be taken to represent the asyzygetic covariants of the degree and order in question.

In order to render the covariants Θ definite as well numerically as in regard to sign, we may suppose that the covariant is in its least terms (i.e. we may reject numerical factors common to all the terms), and we may make the leading term positive. The leading term with the proper numerical coefficient (if different from unity) and with the proper power of x (or the order of the function) annexed, will, when the covariants of a quantic are tabulated, be sufficient to indicate, without any ambiguity whatever, the particular covariant referred to. I subjoin a table of the covariants of a quadric, a cubic and a quartic, and of the covariants of the degrees 1, 2, 3, 4 and 5 respectively of a quintic, and also two other invariants of a quintic.

[Except for the quantic itself, the algebraical sum of the numerical coefficients in any column is = 0, viz. the sum of the coefficients with the sign + is equal to that of the coefficients with the sign –, and I have as a numerical verification inserted at the foot of each column this sum with the sign $\pm$.]

Covariant Tables (Nos. 1 to 26).

No. 1. $(a + 1, b + 2, c + 1 \mid x, y)^2$.

No. 2. $\begin{pmatrix} ac + 1 \\ b^2 - 1 \end{pmatrix} \pm 1$.

The tables Nos. 1 and 2 are the covariants of a binary quadric. No. 1 is the quadric itself; No. 2 is the quadricinvariant, which is also the discriminant.

No. 3. $(a + 1, b + 3, c + 3, d + 1 \mid x, y)^3$.

No. 4. $\begin{pmatrix} ac + 1 & ad + 1 & bd + 1 \\ b^2 - 1 & bc - 1 & c^2 - 1 \end{pmatrix} \mid x, y, \pm 1, \pm 1, \pm 1$.

No. 5. $\begin{pmatrix} a^2d + 1 & abd + 3 & acd - 3 & ad^2 - 1 \\ abc - 3 & ac^2 - 6 & b^2d + 6 & bcd + 3 \\ b^3 + 2 & b^2c + 3 & bc^2 - 3 & c^3 - 2 \end{pmatrix} \mid x, y, \pm 3, \pm 6, \pm 6, \pm 3$.

No. 6. $\begin{pmatrix} a^2d^2 + 1 \\ abcd - 6 \\ ac^3 + 4 \\ b^3d + 4 \\ b^2c^2 - 3 \end{pmatrix} \pm 7$.

The tables Nos. 3, 4 and 5 are the covariants of a binary cubic. No. 3 is the cubic itself; No. 4 is the quadricovariant, or Hessian; No. 5 is the cubicovariant; No. 6 is the invariant, or discriminant. And if we write No. 3 = U , No. 4 = H , No. 5 = Φ , No. 6 = ∇ , then identically,

$$\Phi^2 - \nabla U^2 + 4H^3 = 0.$$

No. 7. $(a + 1, b + 4, c + 6, d + 4, e + 1 \mid x, y)^4$.

No. 8. $\begin{pmatrix} ae + 1 \\ bd - 4 \\ c^2 + 3 \end{pmatrix} \pm 4$.

No. 9. $\begin{pmatrix} ac + 1 & ad + 1 & ae + 1 & bd + 1 \\ b^2 - 1 & bc - 2 & c^2 - 2 & cd - 2 \\ \pm 1 & \pm 2 & \pm 3 & \pm 2 \end{pmatrix} + (be + 2, ce - 1) \mid x, y^4$.

No. 10. (Cubinvariant of the quartic.) $\begin{pmatrix} ace + 1 & a^2d + 1 & d^2e + 1 & abe + 5 & ace & ade - 5 \\ & abc - 3 & abd + 2 & & & \\ ad^2 - 1 & b^4 + 2 & & acd - 15 & ad^2 - 10 & bce + 15 \\ & & ac^2 - 9 & b^3d + 10 & b^4e + 10 & bc^2 - 10 \\ b^2e - 1 & \pm 3 & b^3c + 6 & & & \\ & & \pm 9 & bc^2 & d & c^2d \end{pmatrix}$,

sums $\pm 15, \pm 10, \pm 15, \pm 3$.

[Table No. 10 is rendered schematically; the full layout in the original is a four-column array of literal terms in a, b, c, d, e of degree 3 and order 6 (the cubicovariant of the quartic), with $\pm$ -sum row $\pm 15, \pm 10, \pm 15, \pm 3$.]

No. 11. (Cubicovariant of the quartic.) Eight-column literal table of degree 3 terms, with column sums $\pm 3, \pm 9, \pm 15, \pm 5, \pm 15, \pm 9, \pm 3$.

No. 12. (Discriminant of the quartic.)

$$\begin{aligned}\nabla = & a^2e^3 + 1 \cdot a^2bde^2 \cdot (-12) + a^2c^2e \cdot (-18) + a^2cd^2e \cdot 54 + a^2d^4 \cdot (-27) \\ & + ab^2ce^2 \cdot 54 + ab^2d^2e \cdot (-6) + abc^2de \cdot (-180) + abcd^3 \cdot 108 \\ & + ac^4e \cdot 81 - ac^3d^2 \cdot 54 - b^4e^2 \cdot 27 + b^3cde \cdot 108 \\ & - b^3d^3 \cdot 64 - b^2c^3e \cdot 54 + b^2c^2d^2 \cdot 36 + bc^4d \cdot \dots \\ & - c^6 \cdot \dots,\end{aligned}$$

column sum ± 442 .

The tables Nos. 7, 8, 9, 10 and 11 are the irreducible covariants of a quartic. No. 7 is the quartic itself; No. 8 is the quadrinvariant; No. 9 is the quadricovariant, or Hessian; No. 10 is the cubinvariant; and No. 11 is the cubicovariant. The table No. 12 is the discriminant. And if we write No. 7 = U , No. 8 = I , No. 9 = H , No. 10 = J , No. 11 = Φ , No. 12 = ∇ , then the irreducible covariants are connected by the identical equation

$$JU^3 - IU^2H + 4H^3 + \Phi^2 = 0,$$

and we have

$$\nabla = I^3 - 27J^2.$$

[The Tables Nos. 13 to 24 which follow, and also Nos. 25 and 26 which are given in 143 relate to the binary quintic. I have inserted in the headings the capital letters $A, B, \dots, L$ and also Q and Q' by which I refer to these covariants of the quintic. A is the quintic itself, C is the Hessian, G is the quartinvariant, J a linear covariant; Q is the simplest octinvariant, and Q' is the discriminant. As noticed in the original memoir we have $AI + BF - CE = 0$; and $Q' = G^2 - 128Q$, only the coefficient 128 was by mistake given as 1152.]

A. No. 13. $(a + 1, b + 5, c + 10, d + 10, e + 5, f + 1 \mid x, y)^5$.

B. No. 14.
$$\left(\begin{array}{ccc|c} ae + 1 & af + 1 & bf + 1 & \\ bd - 4 & be - 3 & ce - 4 & \\ c^2 + 3 & cd + 2 & d^2 + 3 & \\ \hline \pm 4 & \pm 3 & \pm 4 & \end{array} \mid x, y \right)^2.$$

C. No. 15.
$$\left(\begin{array}{cccccc|c} ac + 1 & ad + 3 & ae + 3 & af + 1 & bf + 3 & cf + 3 & df + 1 & \\ & bd + 3 & bd + 7 & & ce + 3 & de - 3 & & \\ b^2 - 1 & bc - 3 & be - 6 & cd - 8 & & & c^2 - 1 & \\ & & c^2 - 6 & d^2 - 6 & & & & \\ \hline \pm 1 & \pm 3 & \pm 6 & \pm 8 & \pm 6 & \pm 3 & \pm 1 & \end{array} \mid x, y \right)^6.$$

[Tables D (No. 16), E (No. 17), F (No. 18), G (No. 19), H (No. 20), I (No. 21), J (No. 22), K (No. 23), L (No. 24) follow with the full literal expansions of the corresponding covariants of the binary quintic. The detailed numerical tables are extensive; the column sums (with sign $\pm$) printed at the bottom of each column serve as numerical verifications. The degree–order specifications of D, E, F are $3 \cdot 3, 3 \cdot 5, 3 \cdot 9$; of G, H, I are $4 \cdot 0, 4 \cdot 4, 4 \cdot 6$; of J, K, L are $5 \cdot 1, 5 \cdot 3, 5 \cdot 7$. The column-sum signatures of these tables, where given in the original, are: G : ± 149 ; H : $\pm 33, \pm 54, \pm 33$; I : $\pm 11, \pm 40, \pm 15, \pm 60, \pm 15, \pm 40, \pm 11$; J : $\pm 95, \pm 95$; K : $\pm 57, \pm 129, \pm 129, \pm 57$; L : $\pm 26, \pm 93, \pm 207, \pm 241, \pm 241, \pm 207, \pm 93, \pm 26$.]

No. 25. $Q = +1a^2cdf + \&c.$ [see post, 143.]

No. 26. $Q' = +1a^4f^4 + \&c.$

142.

NUMERICAL TABLES SUPPLEMENTARY TO SECOND MEMOIR
ON QUANTICS.

[Now first published (1889).]

In the present paper I arrange in a more compendious form and continue to a much greater extent the tables (first of each pair) given Nos. 35–39 of my Second Memoir on Quantics, 141, pp. 260–264, which relate to the cubic, the quartic and the quintic functions; and I give the like tables for the sextic, the septimic and the octavic functions respectively. The cubic table exhibits the coefficients of the several terms of the function $1 \div (1 - z \cdot 1 - xz \cdot 1 - x^2z \cdot 1 - x^3z)$, or, what is the same thing, it gives the number of partitions of a given number into a given number of parts, the parts being 0, 1, 2, 3 (repetitions admissible): or again, regarding the letters a, b, c, d , as having the weights 0, 1, 2, 3 respectively, it shows the number of literal terms of a given degree and given weight. And similarly for the quartic, quintic, sextic, septimic and octavic tables respectively, the parts of course being 0, 1, . . . up to 4, 5, 6, 7 or 8, and the letters being $a, b, \dots$ up to e, f, g, h or i . The extent of the tables is as follows:

cubic table	extends to deg-weight	18–27
quartic	„	18–36
quintic	„	18–45
sextic	„	16–45
septimic	„	12–42
octavic	„	10–40

viz. for the quintic, the sextic and the octavic functions these are the deg-weights of the highest invariants respectively. I designate the Tables as the ad -, ae -, af -, ag -, ah - and ai -tables respectively.

It is to be noticed that in the several tables the lower part of each column is for shortness omitted; the column has to be completed by taking into it the series of bottom terms of each of the preceding columns: thus in the af - or quintic table the complete column for degree 3 would be

W	$D\ 3$
0	6
–1	6
–2	5
–3	4
–4	3
–5	2
–6	1
–7	1
–	2
–	1
–	1

where the concluding terms 2, 1, 1 are the bottom terms of the three preceding columns respectively. And the meaning is that for degree 3, and weight 8 or 7, the number of terms is = 6; for weight 7 – 1, = 6, the number of terms is = 6; and similarly for weights 5, 4, 3, 2, 1, the numbers are 5, 4, 3, 2, 1, 1; the numbers are those of the terms

[Figure: Eight-column array (for the quintic, columns labelled $W = 8$ down to $W = 1$) showing the triangle of literal terms in a, b, c, d, e, f of degree 3 broken into columns by weight; each column lists the products a^3 , a^2b , a^2c , . . . down to terms in f , with the count of terms equal to 6, 6, 5, 4, 3, 2, 1, 1 for weights 8, 7, . . . , 1 respectively.]

The like remarks and explanations apply to the other tables.

D	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	W
	1	1	2	3	5	6	8	10	13	15	18	21	25	28	32	36	41	45	50	0
		1	2	3	4	6	8	10	12	15	18	21	24	28	32	36	40	45	50	–1
			1	4	5	7	9	12	14	17	20	24	27	31	35	40	44	49		–2
				1	7	8	11	13	17	19	23	26	31	34	39	43	49			–3
					1	10	12	15	18	22	25	29	33	38	42	47				–4
						1	14	16	20	23	28	31	36	40	46					–5
							1	19	21	26	29	35	38	44						–6
								1	24	27	32	36	42							–7
									1	30	33	39								–8
										1	37									–9

ae-TABLE.

[Figure: Triangular numerical table for the quartic; columns labelled by degree $D = 0, 1, 2, \dots, 18$ and rows labelled by weight W from 0 down to -18 (entries reach weight -36 if the lower triangle were completed). Top row reads 1, 1, 3, 5, 8, 12, 18, 24, 33, 43, 55, 69, 86, 104, 126, 150, 177, 207, 241. Each cell shows the count of literal terms of the indicated degree and weight, with the lower part of each column omitted; the column has to be completed by taking into it the series of bottom terms of the preceding columns.]

af-TABLE.

[Figure: Triangular numerical table for the quintic; columns labelled by degree $D = 0, 1, 2, \dots, 18$ and rows labelled by weight W from 0 down to -27 (entries reach weight -45). Top row reads 1, 1, 3, 6, 12, 20, 32, 49, 73, 102, 141, 190, 252, 325, 414, 521, 649, 795, 967. The structure is the same as for the ad- and ae-tables.]

ag-TABLE.

[Figure: Triangular numerical table for the sextic; columns labelled by degree $D = 0, 1, 2, \dots, 15$ and rows labelled by weight W from 0 down to -30 (entries reach weight -45). Top row reads 1, 1, 4, 8, 18, 32, 58, 94, 151, 227, 338, 480, 676, 920, 1242, 1636.]

ah-TABLE.

[Figure: Triangular numerical table for the septic; columns labelled by degree $D = 0, 1, 2, \dots, 12$ and rows labelled by weight W from 0 down to -30 (entries reach weight -42). Top row reads 1, 1, 4, 10, 24, 49, 94, 169, 289, 468, 734, 1117, 1656.]

ai-TABLE.

[Figure: Triangular numerical table for the octavic; columns labelled by degree $D = 0, 1, 2, \dots, 10$ and rows labelled by weight W from 0 down to -30 (entries reach weight -40). Top row reads 1, 1, 5, 13, 33, 73, 151, 289, 526, 910, 1514.]

The numbers of each table are connected in several ways with those of the preceding tables. One of these connexions, which is of some importance, is best explained by an example: in the *af*-table, 8-20, the number of terms of degree 8 and weight 20 is 73; and we have $73 = 1 + 6 + 16 + 23 + 27$, viz. (see p. 278) those are the numbers of the terms in a^5, a^4, a^3, a^2, a respectively: the complementary factors, (for example) of a^4 are bef^2 , &c. terms in b, c, d, e, f , of the degree 5 and weight 20, and (replacing therein each letter by that which immediately precedes it) these are in number equal to the terms in a, b, c, d, e of the degree 5 and weight $20 - 5 = 15$; thus the number 6 of the terms in question is that for the deg-weight 5-15 of the *ae*-table; and so 1, 6, 16, 23, 27 are the numbers in the *ae*-table for the deg-weights 4-16, 5-15, 6-14, 7-13 and 8-12 respectively, or (making a change rendered necessary by the abbreviated form of the tables) say for the deg-weights 4-0, 5-15, 6-14, 7-13 and 8-12.

143.

TABLES OF THE COVARIANTS M TO W OF THE BINARY QUINTIC: FROM THE SECOND, THIRD, FIFTH, EIGHTH, NINTH AND TENTH MEMOIRS ON QUANTICS.

[Arranged in the present form, 1889.]

The binary quintic has in all (including the quintic itself and the invariants) 23 covariants, which I have represented by the capital letters, $A, B, C, \dots, W$ (alternative forms of two of these are denoted by Q' and S'). The covariants $A, \dots, L$, and also Q, Q' were given in my Second Memoir on Quantics, and except Q and Q' are reproduced in the present reprint thereof, 141; in all these I gave not only the literal terms actually presenting themselves, but also the terms with zero coefficients; in the other covariants however, or in most of them, the terms with zero coefficients were omitted. It is very desirable to have in every case the complete series of literal terms, and in the covariants as here printed they are accordingly inserted: the number of terms is in each case known beforehand by the foregoing af -table, 142, and any omission is thus precluded; by means of this af -table we have the numbers of terms as shown in the following list.

I have throughout (as was done in the Ninth and Tenth Memoirs) expressed the literal terms in a slightly different form from that employed in the Second Memoir: this is done in order to show at a glance in each column the set of terms which contain a given power of a , and in each such set the terms which contain a given power of b .

The numerical verifications are also given not only for the entire column but for each set of terms containing the same power of a ; viz. in most cases, but not always, the positive and negative coefficients of a set have equal sums, which are shown by a number with the sign $\pm$ prefixed. The verification is in some cases given in regard to the subsets involving the same powers of a and b , here also the sums of the positive and negative coefficients are not in every case equal. The cases of inequality will be referred to at the end of this paper.

The whole series of covariants is as follows:

Mem.	No. of table			deg-weight
2	13	$A =$	$(1, 1, 1, 1, 1 \mid x, y)^5$	1 (0...5)
„	14	$B =$	$(3, 3, 3 \mid x, y)^2$	2 (4..6)
„	15	$C =$	$(2, 2, 3, 3, 3, 2, 2 \mid x, y)^6$	2 (2...8)
„	16	$D =$	$(6, 6, 6, 8 \mid x, y)^3$	3 (6..9)
„	17	$E =$	$(5, 6, 6, 6, 6, 5 \mid x, y)^5$	3 (5...10)
„	18	$F =$	$(3, 4, 5, 6, 6, 6, 6, 5, 4, 3 \mid x, y)^9$	3 (3.....12)
„	19	$G =$	$(12 \mid x, y)^0$, Invt.	4–10
„	20	$H =$	$(11, 11, 12, 11, 11 \mid x, y)^4$	4 (8...12)
„	21	$I =$	$(9, 11, 11, 12, 11, 11, 9 \mid x, y)^6$	4 (7.....13)
„	22	$J =$	$(20, 20 \mid x, y)^1$	5 (12, 13)
„	23	$K =$	$(19, 20, 20, 19 \mid x, y)^3$	5 (11..14)
„	24	$L =$	$(16, 18, 19, 20, 20, 19, 18, 16 \mid x, y)^7$	5 (9.....16)
8	83	$M =$	$(32, 32, 32 \mid x, y)^2$	6 (14..16)
„	84	$N =$	$(30, 32, 32, 32, 30 \mid x, y)^4$	6 (13.....17)
9	90	$O =$	$(49, 49 \mid x, y)^1$	7 (17, 18)
„	91	$P =$	$(46, 48, 49, 49, 48, 46 \mid x, y)^5$	7 (15.....20)
2	Q 25 Q' 26	$Q, Q' =$	$(73 \mid x, y)^0$, Invt.	8–20
9	92	$R =$	$(71, 73, 71 \mid x, y)^2$	8 (19..21)
9	S 93	$S, S' =$	$(101, 102, 102, 101 \mid x, y)^3$	9 (21...24)
10	S 93 bis			
9	94	$T =$	$(190, 190 \mid x, y)^1$	11 (27, 28)
3	29	$U =$	$(252 \mid x, y)^0$, Invt.	12–30
9	95	$V =$	$(325, 325 \mid x, y)^1$	13 (32, 33)
5	29a	$W =$	$(967 \mid x, y)^0$, Invt.	18–45

M. No. 83. (Sextic-degree, degree 6 and order 2 covariant of the binary quintic; total of $32 + 57 + 22 = \pm$ -verified column sums for three sets of terms.)

[Figure: The full literal table M is presented as a three-column array. Column 1 lists 32 terms of the form $a^3b^2df^2$, $a^3bcd f^2$, etc., grouped by descending power of a; column 2 lists 57 terms from a different power-of-a subset; column 3 lists 22 terms from a third subset. Column-sum signatures along the bottom (for the three sets and three subsets) are ± 7 , ± 57 , ± 22 for the entire columns, with internal subset sums ± 31 , ± 108 , ± 34 . The full numerical content is preserved in the scanned source.]